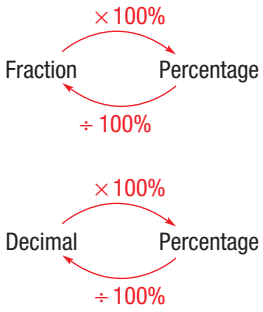
Chapter summary

Financial mathematics

Conversions



Percentage work

15% of 459:

$= \frac{15}{100} \times 459$

Increase 20 by 8%:

$= 20 \times 108\%$

$= 20 \times 1.08$

Decrease 20 by 8%:

$= 20 \times 92\%$

$= 20 \times 0.92$

Profit = selling price − cost price

% Profit = $\frac{\text{profit}}{\text{cost price}} \times 100\%$

% Discount = $\frac{\text{discount}}{\text{cost price}} \times 100\%$

Percentages

'Out of 100'

79 per cent = 79% = $\frac{79}{100}$

Income

Time and a half = 1.5 × hourly rate

Double time = 2 × hourly rate

Commission is x % of total sales.

Simple (flat rate) interest

$I = PRN$

I = simple interest

P = principal ($ invested)

R = rate per year, as a decimal

N = number of years

Compound interest

$A = P(1 + R)^n$

A = final balance

P = principal ($ invested)

R = rate per time period, as a decimal

n = number of time periods

Spreadsheets can be used to manage money or compare investments.

Gross income = total of all money earned

Net income = gross income – deductions

Tax

Taxation = money given from wages (income tax) to the Government

Loans

Balance owing = amount left to repay

Repayment = money given each month to repay the loan amount and the interest

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Chapter review

Chapter tests and worksheets can be found in the Teacher Resource Package. Interactive fill-in-the gaps and drag & drop literacy activities can be found in the Interactive Textbook.

Multiple-choice questions

An online version of this test is available in the Interactive Textbook.

1 28% of \$89 is closest to:

A \$28.00 **B** \$64.08 **C** \$113.92
D \$2492 **E** \$24.92

2 As a percentage, $\frac{21}{60}$ is:

A 21% **B** 3.5% **C** 60%
D 35% **E** 12.6%

3 A price tag says '\$55 (GST inc.)'. How much GST is included in the price?

A \$62.50 **B** \$50 **C** \$49.50
D \$5 **E** \$5.50

4 The gross income for 30 hours at \$15.78 per hour is:

A \$105.78 **B** \$1167.72 **C** \$473.40
D \$747.60 **E** \$74.76

5 If Erin receives \$11 496 commission on the sale of a property worth \$783 200, her rate of commission, to 1 decimal place, was:

A 21% **B** 1.5% **C** 60%
D 15% **E** 12.6%

6 In a given rostered fortnight, Bill works the following number of 8-hour shifts:

- three day shifts (\$21.20 per hour)
- three afternoon shifts (\$24.68 per hour)
- five night shifts (\$33.56 per hour)

His total income for the fortnight is:

A \$305.44 **B** \$2914.68 **C** \$2000
D \$337.68 **E** \$2443.52



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Chapter review

7 A shirt is discounted by 26%. What is the price if it was originally $56?
A $14.56 **B** $41.44 **C** $26.56
D $13.24 **E** $35.22

8 A $5000 loan is repaid by monthly instalments of $200 for 5 years. The amount of simple interest charged is:
A $300 **B** $7000 **C** $12 000
D $2400 **E** $6000

9 The simple interest on $600 at 5% for 4 years is:
A $570 **B** $630 **C** $120
D $720 **E** $30

10 The compound interest on $4600 at 12% p.a. for 2 years is:
A $1104 **B** $5704 **C** $4600
D $5770.24 **E** $1170.24

Short-answer questions

1 Find 15.5% of $9000.

2 Increase $968 by 12%.

3 Decrease $4900 by 7%.

4 The cost price of an item is $7.60. If the mark-up is 50%, determine:
a the retail price
b the profit made

5 An airfare of $7000 is discounted 40% when you fly off-peak. What is the discounted price?

6 A couch is discounted to $375. If this is a 35% discount, find the recommended retail price.



7 Fiona budgets 20% of her income for entertainment. If her yearly income is $37 000, how much could be spent on entertainment in:
a a year?
b a month?
c a week?

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Chapter review

8 Mariah works a 34-hour week at $27.26 per hour. Her net income is 62% of her wage.
- **a** Work out Mariah's net income.
- **b** If 15% of her net income is spent on clothing, determine the amount Mariah can spend each week.
- **c** If Mariah saves $50 each week, what percentage (to 2 decimal places) of her gross weekly income is this?

9 Milan has the following costs to run his car.

hire purchase payment	$350 per month
registration	$685 per year
insurance	$315 per quarter
servicing	$1700 per year
petrol	$90 per week

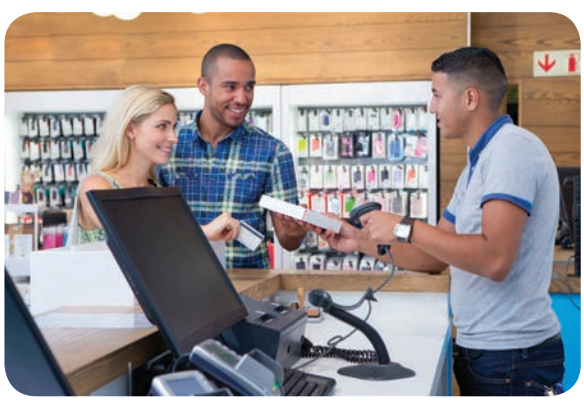
- **a** Find the total cost of running his vehicle for 1 year.
- **b** What percentage (to the nearest per cent) of the overall cost to run the car is the cost of the petrol?

10 Tranh works 36 hours at $28.89 per hour. He pays $142.59 in tax and $22.50 in superannuation. Determine:
- **a** his gross wage
- **b** his net pay

11 Lily receives an annual salary of $47 842. Using the tax table shown, calculate the amount of tax she pays over the year.

Taxable income	Tax on this income
0 – $18 200	Nil
$18 201 – $37 000	19c for each $1 over $18 200
$37 001 – $80 000	$3572 plus 32.5c for each $1 over $37 000
$80 001 – $180 000	$17 547 plus 37c for each $1 over $80 000
$180 001 and over	$54 547 plus 45c for each $1 over $180 000

12 Pedro receives 4.5% commission on sales of $790. Determine the amount of his commission.



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Extended-response questions

1 $5000 is invested at 4% p.a., compounding annually for 3 years.

- **a** What is the value of the investment after the 3 years?
- **b** How much interest is earned in the 3 years?
- **c** How much interest is earned on the investment if it is compounded monthly at 4% p.a. for the 3 years?

2 A vehicle worth $7000 is purchased on a finance package. The purchaser pays 15% deposit and $250 per month over 4 years.

- **a** How much deposit is paid?
- **b** What is the total acount repaid?
- **c** How much interest is paid over the term of the loan?



3 Find the interest paid on a $5000 loan under the following conditions.

- **a** 8% p.a. simple interest over 4 years
- **b** 7% p.a. simple interest over 3 years and 4 months
- **c** 4% p.a. compounded annually over 3 years
- **d** 9.75% p.a. compounded annually over 2 years

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